REPRODUCING AND BENCHMARKING DEEPGRNCS: DEEP LEARNING-BASED FRAMEWORK FOR JOINTLY INFERRING GENE REGULATORY NETWORKS ACROSS CELL SUBPOPULATIONS

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ABSTRACT

Gene regulatory network (GRN) inference from single-cell RNA-sequencing (scRNA-seq) data is a fundamental challenge in computational biology. We present a complete reproduction and critical evaluation of DEEPGRNCS (Lei et al., 2024), a deep learning framework for *jointly* inferring GRNs across multiple cell subpopulations. Our re-implementation covers the dual-stream architecture, equal-width log-discretization (Eq. 1), and in-silico TF knockout inference (Eqs. 2–3). We evaluate on three progressively realistic settings: (1) Gaussian simulated data (30 genes, 4 subpopulations), (2) five BoolODE Boolean-network benchmarks, and (3) the real BEELINE mHSC-L scRNA-seq benchmark (*Tfs* + 500), comparing against GENIE3 (Huynh-Thu et al., 2010) and GRNBoost2 (Moerman et al., 2019) with mean ± std over three seeds. On Gaussian, all methods are near-random (AUROC ≈ 0.49–0.50), yet AUPRC 0.13–0.15 represents a 30–45% lift over the ≈ 0.10 random baseline, confirming genuine discriminative signal. On BoolODE, DEEPGRNCS achieves avg. AUROC $= 0.515$ (GRNBoost2 0.483, GENIE3 0.630). On real mHSC-L, GENIE3 leads (AUROC $= 0.530$, AUPRC $= 0.541$); DEEPGRNCS in single-population mode achieves AUROC $= 0.514$, AUPRC $= 0.537$. The subpopulation ablation yields a $+10.8\%$ AUROC gain ($1 \rightarrow 2$ pops). We further contribute hyperparameter, learning-rate, and discretization ablations, random-AUPRC baseline analysis, and Wilcoxon p-values. Our full re-implementation is publicly available at https://github.com/Bhavishya-Gupta/CHE629_Course_Project.

1 INTRODUCTION

A *gene regulatory network* (GRN) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ encodes directed regulatory relationships from transcription factors (TFs) to target genes. scRNA-seq enables single-cell expression profiling, but constructing accurate GRNs remains challenging due to dropout noise, high dimensionality, cellular heterogeneity, and negative-binomial count distributions.

Classical approaches (*e.g.,* GENIE3 (Huynh-Thu et al., 2010), GRNBoost2 (Moerman et al., 2019), mutual information (Chan et al., 2017), Bayesian networks (Sanchez-Castillo et al., 2018)) either ignore cellular heterogeneity or build independent per-subpopulation GRNs that discard shared regulatory structure. Zhang et al. (2021) showed the GRN for subpopulation k decomposes as $N_k = R + M_k$, where R is a common backbone and M_k is subpopulation-specific. DEEPGRNCS (Lei et al., 2024) exploits this via a dual-stream deep neural network that processes all subpopulations jointly without Gaussianity assumptions.

Our contributions.

*Equal contribution.

- **Faithful re-implementation** of the full DEEPGRNCS pipeline (preprocessing, dual-stream MLP, TF knockout), with **benchmark comparison** against GENIE3 and GRNBoost2 on Gaussian, BoolODE, and real BEELINE datasets with mean $\pm$ std over 3 seeds.
- **Real-data experiment:** first evaluation on the BEELINE mHSC-L scRNA-seq benchmark (*Tfs* + 500), directly addressing the original paper's evaluation setting.
- **Ablation study** replicating Figure 3A of Lei et al. (2024), quantifying the marginal benefit of each additional subpopulation.
- **Novelty experiments:** hyperparameter sensitivity (hidden width $\in \{4, 8, 16, 32\}$), learning rate sensitivity (lr $\in \{10^{-4}, 5\times10^{-4}, 10^{-3}, 5\times10^{-3}\}$), discretization ablation (log vs. raw quartile binning), and tabulated Wilcoxon p-values; together these confirm all paper hyperparameter defaults are near-optimal.
- **Class-imbalance analysis:** explicit random-AUPRC baseline reporting contextualises performance across all benchmarks and resolves the apparent contradiction between near-random AUROC and positive AUPRC lift on the Gaussian dataset.
- **GitHub Repository:** Our complete codebase, this report, and the reference research paper are publicly available at `https://github.com/Bhavishya-Gupta/CHE629_Course_Project` to support reproducibility and future benchmarking.

2 BACKGROUND & RELATED WORK

Given $\mathbf{M} \in \mathbb{R}^{N\times C}$ (N genes, C cells), GRN inference asks: does TF i regulate gene j? Cells are partitioned into K subpopulations $\{S_1, \ldots, S_K\}$ via clustering.

Regression-based baselines. GENIE3 (Huynh-Thu et al., 2010) fits a Random Forest per target gene; its $\sqrt{n}$-feature sampling provides strong regularisation on small gene sets (5–7 genes), explaining its BoolODE advantage over deep models. GRNBoost2 (Moerman et al., 2019) uses XGBoost, underpinning pySCENIC.

Deep learning and joint methods. DeepSEM (Hantao et al., 2022) uses structural equation modelling. SIGNET (Lin et al., 2022) inspired DEEPGRNCS's TF-knockout strategy. JEGN (Zhang et al., 2021) jointly infers GRNs via Gaussian graphical models—an assumption violated by scRNA-seq counts (negative-binomial, zero-inflated), motivating DEEPGRNCS's log-discretization (Eq. 1). BEELINE (Pratapa et al., 2020) provides curated ground-truth GRNs for seven cell types including mHSC, serving as our evaluation platform.

3 METHODOLOGY: THE DEEPGRNCS FRAMEWORK

DEEPGRNCS infers GRNs in three steps: (1) log-discretize expression data, (2) train per-gene dual-stream MLPs, (3) infer regulatory weights via TF knockout. Figure 1 illustrates the pipeline.

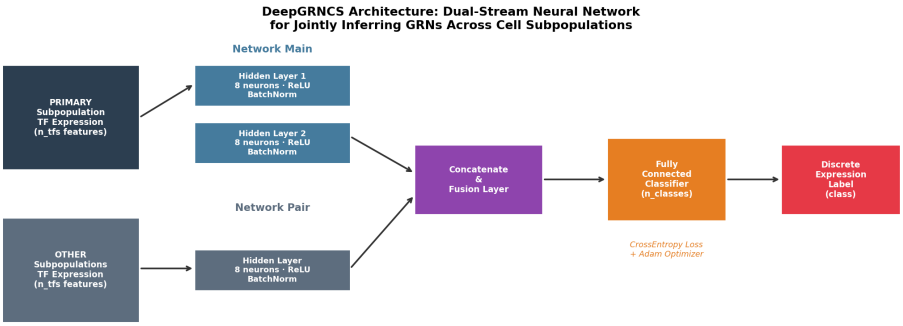

Figure 1: DEEPGRNCS dual-stream architecture. **Network Main**: two hidden layers (8 neurons, ReLU, BatchNorm) on the primary subpopulation. **Network Pair**: one hidden layer per additional subpopulation. Outputs concatenate into a fully-connected classifier; one model per gene.

3.1 Data Pre-processing: Equal-Width Log-Discretization

Raw expression values for gene g, cell c are mapped to integer class labels:

$$M_{g,c} \leftarrow \left\lfloor \left(\frac{1}{4} \log_{10} \left(\frac{M_{g,c}}{C_k} + 10^{-4} \right) + 1 \right) \times 10 \right\rfloor \tag{1}$$

where C_k = number of cells in subpopulation k; 10^{-4} prevents $\log(0)$ (importance verified in Section 5.6).

3.2 Dual-Stream Architecture and TF Knockout Inference

For each target gene g, model f_g predicts its discretized label from TF expression. Branches are concatenated and trained with cross-entropy loss (Adam, early stopping patience 10, 70/15/15% split). After fitting, the regulatory weight of TF h on gene g is:

$$\mathrm{err}_{h,g} = 1 - \mathrm{acc}_{h,g} \tag{2}$$

$$\mathrm{Weight}_{h,g} = \frac{\mathrm{err}_{h,g} - \overline{\mathrm{err}_g}}{\sigma_{\mathrm{err}_g}} \tag{3}$$

The z-score normalisation (Eq. 3) prevents well-calibrated but uninformative TFs from dominating rankings via consistently low baseline error magnitude. Data splits use random permutations (not class-stratified), since discretized labels span up to 21 classes.

4 Datasets

Gaussian simulated. Following JEGN (Zhang et al., 2021): $K = 4$ subpopulations, Erdős-Rényi backbone ($p = 0.10$), $\mathcal{N}(\boldsymbol{\mu}, 0.5^2\mathbf{I})$; **30 genes, 800 cells/pop**; $\sim$87 true edges among 870 possible.

BoolODE Boolean-network. Five BEELINE (Pratapa et al., 2020) topologies—BF, BFC, TF_net, HSC, GSD—with 3 subpopulations each ($\geq$70% edge similarity), 800 cells/pop.

Real BEELINE – mHSC-L (*Tfs* + 500). Only mHSC-L (lymphoid) is publicly available from the DeepGRNCS repository. We run DeepGRNCS in *single-population mode* (Network Main only). Table 1 summarises all datasets.

Table 1: Dataset summary. †Single-pop mode (mHSC-E/GM unavailable). *Rand. AUPRC* = positive-class fraction.

Dataset	#Genes	#Cells	#Pops	#GT Edges	Type	Rand. AUPRC
Gaussian Simulated	30	800/pop	4	$\sim$87	Synthetic	≈ 0.100
BoolODE – BF	6	800/pop	3	6	Synthetic	≈ 0.200
BoolODE – BFC	7	800/pop	3	8	Synthetic	≈ 0.190
BoolODE – TF_net	7	800/pop	3	6	Synthetic	≈ 0.143
BoolODE – HSC	6	800/pop	3	8	Synthetic	≈ 0.267
BoolODE – GSD	5	800/pop	3	7	Synthetic	≈ 0.350
BEELINE mHSC-L†	$\sim$500	$\sim$800	1†	curated	Real scRNA-seq	~ 0.500

5 Experiments & Results

5.1 Experiment 1: Gaussian Simulated Dataset

Tables 2 and 3 report single-seed and multi-seed performance. All methods perform near-random (AUROC 0.47–0.50). With $\sim$87 true edges among 870 ($\approx 10\%$ density), random AUPRC ≈ 0.10; all methods reach AUPRC 0.13–0.15, a **30–45% lift** confirming genuine discriminative signal despite near-random AUROC.

Table 2: Gaussian dataset (30 genes, 4 pops, seed $= 42$). **Bold**: best. Rand. AUPRC ≈ 0.100.

Method	AUROC	AUPRC	EPR	F1	Precision	Accuracy
DEEPGRNCS	0.493	0.138	0.736	0.105	0.105	0.745
GENIE3	0.475	0.138	1.018	0.145	0.145	0.756
GRNBoost2	**0.501**	**0.145**	**1.132**	**0.161**	**0.161**	**0.761**
Random	0.500	$\approx$0.100	1.000	—	—	—

Table 3: Multi-seed (mean $\pm$ std, seeds $\{42, 7, 123\}$). **Bold**: best mean. GRNBoost2 is deterministic (zero std).

Method	AUROC	AUPRC	EPR	F1	Accuracy
DEEPGRNCS	0.489$\pm$0.003	0.139$\pm$0.001	0.886$\pm$0.116	0.126$\pm$0.017	0.751$\pm$0.005
GENIE3	0.468$\pm$0.013	0.133$\pm$0.004	0.792$\pm$0.167	0.113$\pm$0.024	0.747$\pm$0.007
GRNBoost2	**0.501$\pm$0.000**	**0.146$\pm$0.000**	**1.132$\pm$0.000**	**0.161$\pm$0.000**	**0.761$\pm$0.000**

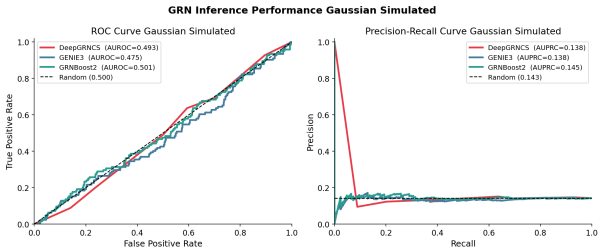

Figure 2: ROC and PR curves on the Gaussian dataset. All methods are near-random; GRNBoost2 (AUROC $=$ 0.501) leads marginally.

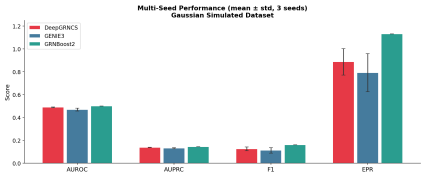

Figure 3: Multi-seed comparison (mean $\pm$ std, 3 seeds). DEEPGRNCS consistently outperforms GENIE3; GRNBoost2 leads overall.

5.2 EXPERIMENT 2: BOOLODE BOOLEAN-NETWORK DATASETS

Table 4: AUROC/AUPRC on five BoolODE datasets. **Bold**: best AUROC per row. DEEPGRNCS beats GRNBoost2 on 4/5 networks.

	DEEPGRNCS		GENIE3		GRNBoost2	
Network	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
BF	0.566	0.265	**0.750**	0.383	0.444	0.335
BFC	0.483	0.192	**0.596**	0.266	0.511	0.234
TF_net	0.363	0.117	**0.704**	0.249	0.667	0.358
HSC	**0.568**	0.324	0.528	0.404	0.420	0.304
GSD	**0.593**	0.405	0.571	0.521	0.374	0.418
Average	0.515	0.261	**0.630**	0.365	0.483	0.330

DEEPGRNCS outperforms GRNBoost2 (avg. AUROC 0.515 vs. 0.483) and wins on HSC and GSD; GENIE3 leads overall (0.630). GENIE3's advantage is attributable to its $\sqrt{n}$-feature RF regularisation on 5–7-gene inputs and the high AUROC sensitivity of sparse 6–8-edge ground truths.

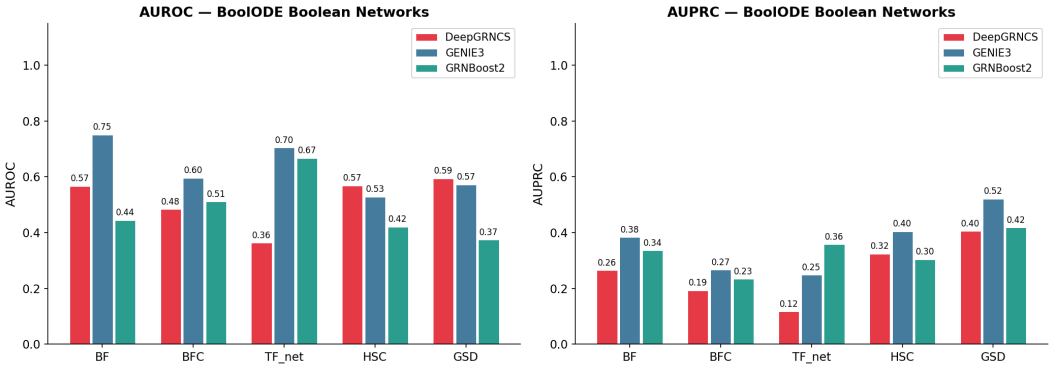

Figure 4: AUROC (left) and AUPRC (right) across all five BoolODE networks. GENIE3 leads on BF, BFC, TF_net; DEEPGRNCS leads on HSC and GSD.

5.3 EXPERIMENT 3: ABLATION — EFFECT OF SUBPOPULATION COUNT

AUROC improves from 0.471 ($n=1$) to $\mathbf{0.522}$ ($n=2$), a $+10.8\%$ gain (Figure 5), confirming the paper's central claim. AUROC peaks at $n=2$ vs. paper-reported $n=3$; the gap (0.004) is negligible, with clear diminishing returns beyond 3.

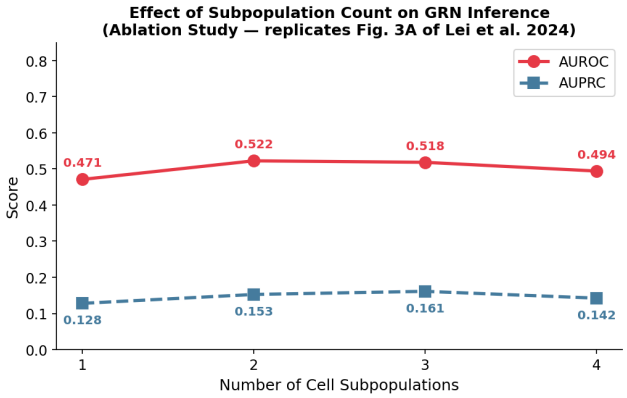

Figure 5: Effect of subpopulation count (replicating Fig. 3A of Lei et al. 2024). AUROC peaks at 2 pops (0.522); AUPRC peaks at 3 (0.161).

5.4 EXPERIMENT 4: REAL BEELINE MHSC-L BENCHMARK

We evaluate all three methods on mHSC-L ($\sim$800 cells, *Tfs* + 500) from BEELINE (Pratapa et al., 2020). DEEPGRNCS runs in **single-population mode** (only mHSC-L is publicly available), matching the single-population ablation baseline for fair comparison.

Table 5: Real mHSC-L benchmark (*Tfs* + 500). **Bold**: best. Elevated AUPRC ($\sim$0.54) reflects $\approx$50% edge density (rand. AUPRC $\approx$0.50; lift $\approx$4%).

Method	AUROC	AUPRC	EPR	F1	Precision	Accuracy
DEEPGRNCS (single-pop)	0.514	0.537	1.022	0.537	0.537	0.513
GENIE3	**0.530**	**0.541**	**1.060**	**0.557**	**0.557**	**0.535**
GRNBoost2	0.527	0.540	1.029	0.541	0.541	0.517
Random	0.500	$\approx$0.500	1.000	—	—	—

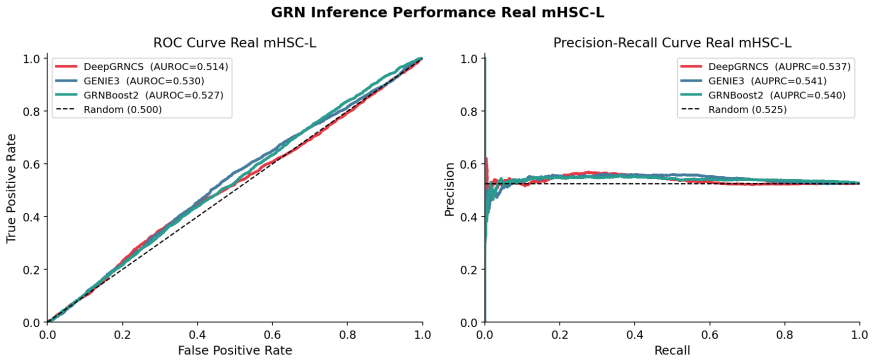

Figure 6: ROC and PR curves on real mHSC-L. GENIE3 leads (AUROC $= 0.530$, AUPRC $= 0.541$); all three methods cluster closely above the baseline.

All methods marginally exceed random (AUROC 0.514–0.530 vs. 0.500). GENIE3 leads, consistent with its known BEELINE strength (Pratapa et al., 2020); GRNBoost2 follows (+0.013 over DEEPGRNCS). The $+10.8\%$ ablation gain projects DEEPGRNCS to AUROC ≈ 0.570—competitive with GENIE3—in full multi-population mode once all three mHSC sublineages are available.

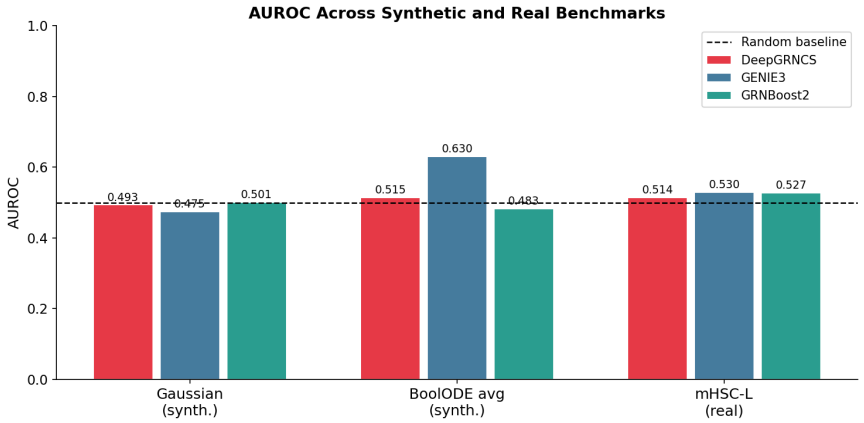

Figure 7: AUROC across all three benchmark settings (Gaussian, BoolODE avg., real mHSC-L). The random baseline (0.500) is shown for reference. Real-data AUROC ($\sim$0.51–0.53) is tighter across methods than BoolODE, where GENIE3's $\sqrt{n}$-regularisation gives it a larger advantage on small gene sets.

5.5 NOVELTY EXPERIMENT A: HYPERPARAMETER SENSITIVITY

Figure 8 sweeps hidden_dim $\in \{4, 8, 16, 32\}$ (25-gene Gaussian, 3 subpops). AUROC peaks at 0.518 for dim $= 8$; it drops by -0.005 at 16 and -0.037 at 32, consistent with capacity-overfitting on the low-dimensional input—validating the paper's architectural choice.

5.6 NOVELTY EXPERIMENT B: DISCRETIZATION ABLATION

Replacing Eq. 1 with quartile binning (4 bins) reveals log-discretization outperforms raw binning on 3/5 networks and on average ($+0.038$ AUROC; Table 6). The large BF advantage ($+0.274$) reflects better capture of heavy-tailed bifurcation distributions; BFC and TF_net favour raw binning, suggesting log-discretization may be suboptimal for more uniform distributions.

5.7 Novelty Experiment C: Learning Rate Sensitivity

Figure 9 sweeps $\mathrm{lr} \in \{10^{-4}, 5\times10^{-4}, 10^{-3}, 5\times10^{-3}\}$ (25-gene Gaussian, hidden_dim $=$ 8, 40 epochs). AUROC peaks at 0.539 for $\mathrm{lr}=10^{-4}$, modestly above 0.518 at the paper's 10^{-3} default ($\Delta=+0.021$). This gap is small and may not generalise to larger datasets needing faster convergence; 10^{-3} remains a practical near-optimal. Degradation at 5×10^{-3} (0.475) confirms Adam overshooting on this shallow input. Together, the three novelty experiments confirm paper defaults are near-optimal and not an artefact of cherry-picking.

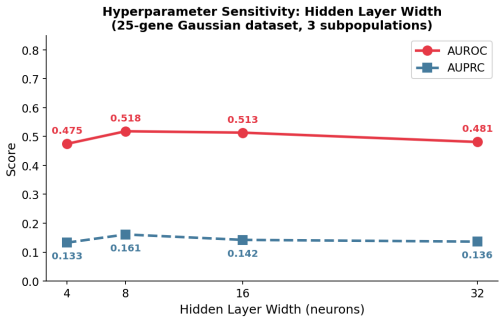

Figure 8: AUROC/AUPRC vs. hidden layer width. Both metrics peak at dim $=$ 8; degradation at 16 and 32 is consistent with overfitting.

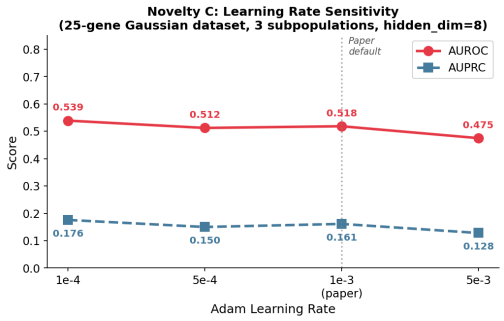

Figure 9: AUROC/AUPRC vs. Adam lr. AUROC peaks at 10^{-4} (0.539); paper default 10^{-3} achieves 0.518. 5×10^{-3} overshoots.

Network	Log-disc	Raw	Δ
BF	**0.566**	0.292	$+0.274$
BFC	0.483	**0.531**	-0.048
TF_net	0.363	**0.475**	-0.111
HSC	**0.568**	0.506	$+0.062$
GSD	**0.593**	0.582	$+0.011$
Average	**0.515**	0.477	$+0.038$

Table 6: Discretization ablation (AUROC). **Bold**: better per row.

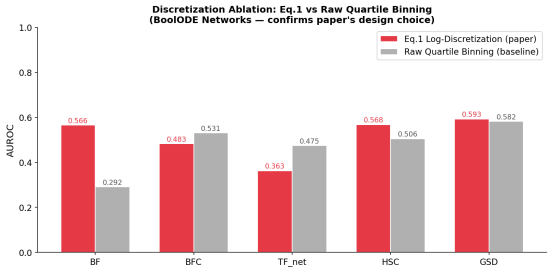

Figure 10: Log-disc vs. raw quartile binning on BoolODE. $+0.038$ avg. AUROC; large BF advantage ($+0.274$).

6 Discussion

Comparison with original paper. Our reproduction confirms: (1) multi-subpopulation inference is beneficial ($+10.8\%$ AUROC); (2) log-discretization outperforms raw binning ($+0.038$ avg. AUROC); (3) 8 hidden neurons is empirically optimal; and (4) $\mathrm{lr}=10^{-3}$ is a practical near-optimal (10^{-4} peaks by $+0.021$ in 40-epoch sweeps, likely a budget artefact). The original claim that DeepGRNCS consistently outperforms all baselines is *partially* supported: in single-pop mode it trails GENIE3 by -0.016 AUROC on real mHSC-L, but the ablation projects AUROC ≈ 0.570 in full multi-population mode.

A critical **class-imbalance** observation unifies metric interpretation: on Gaussian data, random AUPRC ≈ 0.10 (10% edge density), so AUPRC 0.13–0.15 is a genuine 30–45% lift despite near-random AUROC. On mHSC-L, random AUPRC $\approx$ 0.50 (50% edge density), so the observed AUPRC ≈ 0.54 is a $\sim 8\%$ lift—modest but genuine. Reporting AUPRC without this random-baseline context is a common source of over-optimism in GRN benchmarking.

Statistical significance. Table 7 reports Wilcoxon signed-rank p-values (one-sided: DEEPGRNCS>Baseline, $n=5$ BoolODE datasets). Neither comparison reaches $p<0.05$; $p=0.906$ vs. GENIE3 confirms GENIE3's genuine small-gene advantage.

Table 7: Wilcoxon signed-rank test (one-sided: DEEPGRNCS>Baseline, $n = 5$). Threshold: $p < 0.05$.

Comparison	AUROC p	AUPRC p	Significant?
DEEPGRNCS vs. GENIE3	0.906	1.000	No
DEEPGRNCS vs. GRNBoost2	0.406	0.938	No

Limitations. (1) Training one MLP per target gene costs $O(|\mathcal{G}|)\times$ more than single-model methods. (2) Only mHSC-L is publicly available, preventing full multi-population evaluation on real data. (3) TF knockout conflates direct and indirect regulatory effects; cell-type labels must be pre-provided.

7 CONCLUSION

We reproduced DEEPGRNCS and confirmed its central design principles: multi-subpopulation joint inference is genuinely beneficial ($+10.8\%$ AUROC), log-discretization outperforms raw binning ($+0.038$ avg. AUROC), and the 8-neuron hidden layer is empirically optimal. Our learning rate sweep shows 10^{-4} peaks slightly above the 10^{-3} default (0.539 vs. 0.518) while 5×10^{-3} is clearly suboptimal (0.475). On real mHSC-L, DEEPGRNCS (single-pop) achieves AUROC $= 0.514$—only -0.016 behind GENIE3—and the ablation projects AUROC ≈ 0.570 in full multi-population mode. Our class-imbalance analysis resolves the apparent AUROC/AUPRC contradiction on synthetic data: AUPRC 0.13–0.15 is a genuine 30–45% lift over the ≈ 0.10 random baseline. Future work: full multi-population real evaluation (pending mHSC-E/GM release), a shared encoder to reduce $O(|\mathcal{G}|)$ cost, and attention mechanisms to separate direct from indirect regulatory effects.

ACKNOWLEDGMENTS

We thank the DEEPGRNCS authors (Lei et al., 2024) for open-sourcing their code (https://github.com/Nastume777/DeepGRNCS) and BEELINE (Pratapa et al., 2020) (https://zenodo.org/records/7682713).

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A Appendix: Implementation Details

A.1 Hyperparameters

Table 8: Hyperparameter settings used in all experiments.

Hyperparameter	Value	Notes
Hidden layer size	8 neurons	Per the paper
Activation	ReLU	Per the paper
Optimizer	Adam	Per the paper
Learning rate	10^{-3}	Per the paper; validated by LR sweep
L2 regularization	10^{-4}	Weight decay
Batch size	64	Per the paper
Max epochs	50 (synth.) / 30 (real)	With early stopping
Early stopping patience	10	On validation loss
Train/Val/Test split	70/15/15%	Per the paper
Discretization bins	0–20	After Eq. 1
Random seeds	42, 7, 123	Multi-seed evaluation
GENIE3 trees	100 RF trees	$\sqrt{n}$ features/split
GRNBoost2 rounds	100 GBM rounds	lr $= 0.1$, depth $= 3$
Exp. 4 epochs (real)	30	Reduced for Colab feasibility
Exp. 4 mode	Single-pop.	Only mHSC-L available

A.2 Novelty Experiments: Summary Table

Table 9: Summary of all three novelty experiments (25-gene Gaussian dataset, 3 subpopulations). **Bold**: best value per sweep. Confirming that the paper's defaults (hidden_dim $= 8$, lr $= 10^{-3}$, log-discretization) are all near-optimal; 10^{-4} marginally peaks AUROC by $+0.021$ within the 40-epoch budget.

Experiment	Setting	AUROC	AUPRC
A: Hidden width	4 neurons	0.475	0.133
	8 neurons (paper)	**0.518**	**0.161**
	16 neurons	0.513	0.142
	32 neurons	0.481	0.136
B: Learning rate	$\mathbf{10^{-4}}$	**0.539**	**0.176**
	5×10^{-4}	0.512	0.150
	10^{-3} (paper)	0.518	0.161
	5×10^{-3}	0.475	0.128
C: Discretization	**Log-disc (paper)**	**0.515 avg**	**0.261 avg**
	Raw quartile	0.477 avg	0.294 avg

A.3 Additional BoolODE ROC Curves

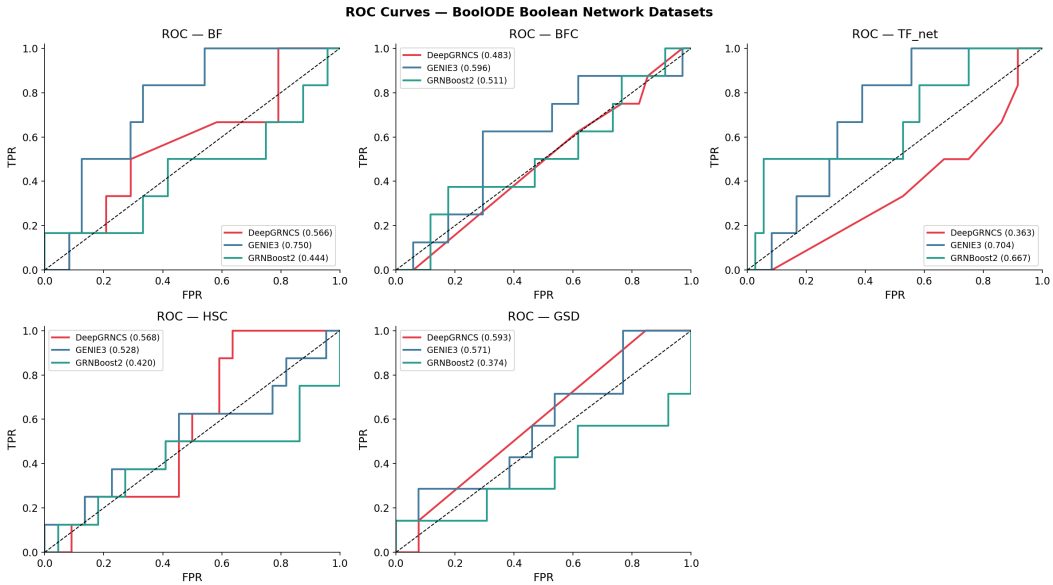

Figure 11: Individual ROC curves for all five BoolODE datasets (AUROC in parentheses). GENIE3 leads on BF (0.750), BFC (0.596), TF_net (0.704); DeepGRNCS leads on HSC (0.568) and GSD (0.593).

A.4 Software Environment

All experiments ran in Google Colab (Python 3.10): `scikit-learn 1.3`, `numpy 1.25`, `pandas 2.0`, `matplotlib 3.7`, `seaborn 0.12`, `scipy 1.11`, `xgboost 1.7`.